Power Electronics I Project

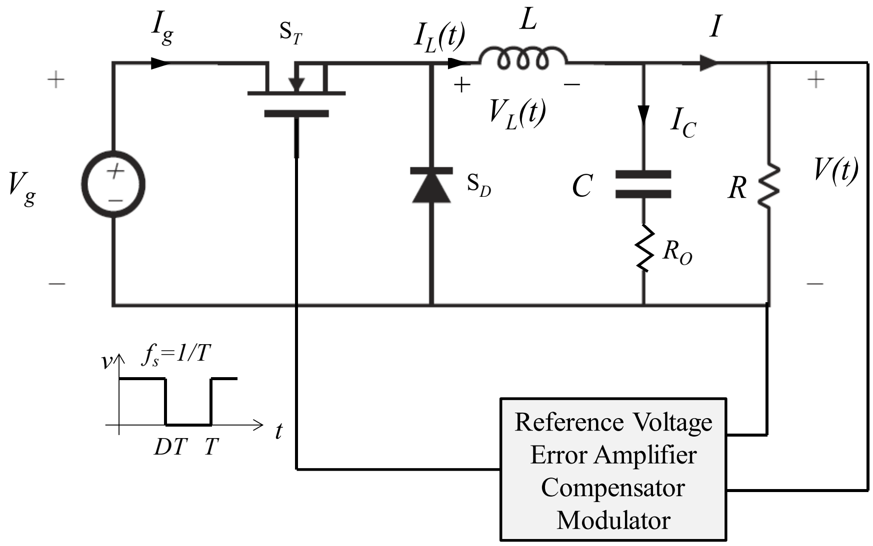
(Due: Dec 3, 2025)

The purpose of this project is to design the feedback control for the 75 W Buck converter shown in Fig. 1 with the following nominal parameters:

$V_g = 35\text{V}$, $V = 12\text{ V} \pm 1\ \%$, $L = 100\ \mu\text{H}$, $R_O = 10\text{m}\,\Omega$,

$C = 47\ \mu\text{F}$, $R = 3\ \Omega$, $V_M = 1\text{ V}$, $f_s = 150\text{ kHz}$.

Where V_M is the saw-tooth PWM modulator's voltage amplitude. Reference voltage $V_{ref} = 5\text{ V}$.



The design specifications are:

- **Loop-gain cross-over frequency: f_c** (please design for the maximum f_c that you can achieve)**,**
- **Phase margin: $\varphi_m \geq 30°$,**

Please design the complete feedback circuit including a PID compensator, voltage sensing, and the values of all components except for the modulator implementation.

Your design should follow the steps below:

Step 1: (2 point)

Find steady state parameters

Step 2: (3 points)

Derive open loop small signal G_{vd}, G_{vg} and Z_o, find quality factor Q, zero frequency ω_Z, pole frequency ω_P and coefficient G_o, etc

Step 3: (3 points)

Draw bode plots for G_{vd}, G_{vg}, Z_o and loop gain T in Matlab or other software

Step 4: (3 points)

Compensator $G_c(s)$ and voltage sensing H(s) design based on the bode plot of loop gain, modulator's gain and the requirements of f_c and Φ_m

Step 5: (2 point)

Draw a bode plot for the loop gain with the designed compensator $G_c(s)$ plugged in in Matlab or other software which you prefer

Step 6: (2 point)

Compensator implementation with an operational amplifier

Step 7: (2 point)

The final circuit including a voltage sensing, an error amplifier, a compensator, and a modulator.

Step 8 **(**3 point, undergraduate 3 + 1 bonus point**):**

Close loop circuit simulation verification for your design in time domain (please show simulated time domain response waveforms for V_{out} when: 1) input voltage V_g steps from 48V to 60V, 2) load current steps from 4A to 3A.